

Assignment 2: Discrete Random Variables (Total Marks 60)

Problem 1: Chess Match Duration

4 Marks

Fischer and Spassky play a chess match in which the first player to win a game wins the match. If 10 successive games are drawn, the match is declared a draw.

Each game results in:

- Fischer wins with probability 0.4,
- Spassky wins with probability 0.3,
- Draw with probability 0.3,

independently of all previous games.

- (a) What is the probability that Fischer wins the match?
- (b) Let T denote the duration of the match (number of games played). Find the probability mass function (PMF) of T .

Problem 2: Birthday Problem (Departmental Gathering)

4 Marks

In Probability class, you attend a departmental gathering with 500 guests (excluding yourself).

Assume that birthdays are uniformly distributed over 365 days (ignore February 29), and that guests' birthdays are independent.

- (a) What is the probability that **exactly one** other guest has the same birthday as you? Compute this probability exactly.
- (b) Approximate the probability in part (a) using a Poisson approximation. Clearly state the parameter used in the Poisson PMF.

Problem 3: Internet Service Provider Model

6 Marks

An internet service provider operates 50 modems to serve 1000 customers.

At any given time, each customer independently requires a connection with probability 0.01.

- (a) Let X be the number of customers requiring a connection at a given time. Find the exact PMF of X .
- (b) Approximate the PMF of X using a Poisson approximation. Clearly specify the Poisson parameter.
- (c) What is the probability that more customers require a connection than there are modems available? Provide both:
 - the exact expression, and
 - the Poisson approximation.

Problem 4: Door's Key

4 Marks

You just rented a large house and the realtor gave you 5 keys, one for each of the 5 doors of the house. Unfortunately, all keys look identical, so to open the front door, you try them at random.

(a) Find the **PMF** of the number of trials you will need to open the door, under the following alternative assumptions: (1) after an unsuccessful trial, you mark the corresponding key, so that you never try it again, and (2) at each trial you are equally likely to choose any key.

(b) Repeat part (a) for the case where the realtor gave you an extra duplicate key for each of the 5 doors.

Problem 5: Four Independent Die

2 Marks

Consider four independent rolls of a 6-sided die. Let X be the number of 1's and let Y be the number of 2's obtained. What is the joint PMF of X and Y ?

Problem 6: Expectation, Mean, and Variance

8 Marks

Let X be a random variable with PMF

$$p_X(x) = \begin{cases} \frac{x^2}{a}, & x = -3, -2, -1, 0, 1, 2, 3, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find a and $E[X]$.
- (b) Let $Z = (X - E[X])^2$. Find the PMF of Z .
- (c) Using part (b), compute $\text{Var}(X)$.
- (d) Compute $\text{Var}(X)$ using the formula

$$\text{Var}(X) = \sum_x (x - E[X])^2 p_X(x).$$

Problem 7: Mod PMF

4 Marks

Let X be a random variable that takes values $0, 1, 2, \dots, 9$ with equal probability $1/10$.

- (a) Find the PMF of the random variable $Y = X \bmod 3$.
- (b) Find the PMF of the random variable $Y = 5 \bmod (X + 1)$.

Problem 8: Stock Market Trader

4 Marks

A stock market trader buys 100 shares of stock A and 200 shares of stock B .

Let X and Y be the price changes of A and B , respectively, over a certain time period. Assume that the joint PMF of X and Y is uniform over the set of integers x and y satisfying

$$-2 \leq x \leq 4, \quad -1 \leq y - x \leq 1.$$

- (a) Find the marginal PMFs and the means of X and Y .
- (b) Find the mean of the trader's total profit.

Problem 9: Traffic Ticket

8 Marks

You drive to work 5 days a week for 50 weeks (250 days total). On any given day, you get a traffic ticket with probability $p = 0.02$, independently of other days.

Let X be the total number of tickets you receive during the year.

- (a) What is the probability that the number of tickets you get is exactly equal to the expected value of X ?
- (b) Approximate the probability in part (a) using a Poisson approximation.

- (c) Each ticket is \$10, \$20, or \$50 with respective probabilities 0.5, 0.3, and 0.2, independently of other tickets. Find the mean and variance of the total amount of money paid in tickets during the year.
- (d) Suppose you do not know the probability p of getting a ticket. If you received 5 tickets during the year, estimate p using the sample mean:

$$\hat{p} = \frac{5}{250}.$$

Problem 10: Email on Weekday

4 Marks

The number of emails that I get on a weekday (Monday through Friday) can be modeled by a Poisson distribution with an average of 16 emails per minute. The number of emails that I receive on weekends (Saturday and Sunday) can be modeled by a Poisson distribution with an average of 130 emails per minute.

1. What is the probability that I get no emails in an interval of length 4 hours on a Sunday?
2. A random day is chosen (all days of the week are equally likely to be selected), and a random interval of length one hour is selected on the chosen day. It is observed that I did not receive any emails in that interval. What is the probability that the chosen day is a weekday?

Problem 11: Urn's Probability

4 Marks

An urn consists of 20 red balls and 30 green balls. We choose 10 balls at random from the urn. The sampling is done without replacement (repetition not allowed).

1. What is the probability that there will be exactly 4 red balls among the chosen balls?
2. Given that there are at least 3 red balls among the chosen balls, what is the probability that there are exactly 4 red balls?

Problem 12: (St. Petersburg Paradox)

8 Marks

A casino offers a game of chance for a single player in which a fair coin is tossed at each stage. The pot starts at 1 dollar and is doubled every time a head appears. The first time a tail appears, the game ends and the player wins whatever is in the pot. Thus the player wins 1 dollar if a tail appears on the first toss, 2 dollars

if a head appears on the first toss and a tail on the second, 4 dollars if a head appears on the first two tosses and a tail on the third, 8 dollars if a head appears on the first three tosses and a tail on the fourth, and so on. In short, the player wins 2^{k-1} dollars if the coin is tossed k times until the first tail appears. What would be a fair price to pay the casino for entering the game?

1. Let X be the amount of money (in dollars) that the player wins. Find $E[X]$.
2. What is the probability that the player wins more than 65 dollars?
3. Now suppose that the casino only has a finite amount of money. Specifically, suppose that the maximum amount of money that the casino will pay you is 2^{30} dollars (around 1.07 billion dollars). That is, if you win more than 2^{30} dollars, the casino is going to pay you only 2^{30} dollars. Let Y be the money that the player wins in this case. Find $E[Y]$.